

Network Scanning for Ethical Hacking & Cybersecurity

Understanding ports, services, vulnerabilities, and the tools used to identify attack surfaces.

◆ What is Network Scanning?

Network scanning is the process of **identifying active hosts, open ports, running services, and potential vulnerabilities** within a network. It helps ethical hackers understand *where* a system is exposed and what entry points malicious attackers may exploit.

- ✓ Used by Ethical Hackers
- ✓ Used by SOC, VAPT teams
- ✓ Used in vulnerability management

Network scanning is essential because **every network—internal or internet-facing—has weak points**. Your job is to find them before attackers do.

◆ Reconnaissance vs Network Scanning

Many beginners confuse **Reconnaissance** with **Scanning**, but they serve completely different purposes in hacking.

◆ Reconnaissance = Information Gathering

This is about understanding your target:

- IP ranges
- Domain names
- Technologies used
- Publicly available data

- Employee info
- Metadata
- Website structure

Goal: Build a profile of the target.

◆ **Network Scanning = Identifying Entry Points**

Once you understand the target, scanning helps you find **where** you can attack.

- Live hosts
- Open ports
- Running services
- OS types
- Vulnerabilities
- Weak configurations

Goal: Map attack surfaces and infiltration points.

◆ **Types of Network Scanning**

Network scanning is broadly divided into two categories:

1 Port Scanning

Finds open, closed, and filtered ports to identify:

- Running services
- Entry points
- Weak configurations

2 Vulnerability Scanning

Finds:

- Misconfigurations
- Unpatched systems
- Weak protocols

- Known CVEs

Let's explore both in depth.

◆ **Port Scanning**

Port scanning helps identify how a device communicates with the outside world.

An attacker first needs to find:

- Which services are running
- Which ports are open
- Which ports are filtered
- Which ports are vulnerable

To do this, different scanning techniques are used.

◆ **Common Types of Port Scanning**

Here's a breakdown of the most used scanning techniques and what they reveal:

✓ **TCP Connect Scan**

- Completes full TCP handshake
- Easiest to detect
- Reliable but noisy

✓ **SYN Scan (Half-Open Scan)**

- Sends SYN → waits for SYN/ACK
- Does not complete handshake
- Most stealthy, fastest
- Widely used by ethical hackers

✓ **UDP Scan**

- Identifies UDP-based services (DNS, SNMP, DHCP)

- Slow but necessary

✓ **ACK Scan**

- Used to map firewall rules
- Detects filtered vs unfiltered ports

✓ **Window Scan**

- Uses TCP window size to infer port state

✓ **FIN Scan**

- Sends FIN packet to check closed/open ports
 - Useful when firewalls block SYN scans
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Vulnerability Scanning

Vulnerability scanning checks for:

- Poor coding
- Misconfigurations
- Weak services
- Outdated software
- CVE exposure

Most vulnerability scanners compare results against massive vulnerability databases.

Common Vulnerability Types Found:

- **Open directory listings**
- **Old Apache/Nginx versions**
- **Default credentials**
- **Open database ports**
- **Weak SSL/TLS configurations**
- **Old PHP versions**

- **SMB vulnerabilities**
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◆ **Practical Network Scanning Tools & How to Use Them**

Below are the most important tools you should know (from beginner to expert).

◆ **Nmap (Network Mapper)**

Nmap is the **king of scanning tools**—powerful, flexible, and included in Kali Linux.

▶ **Basic Scan**

```
nmap 1.2.3.4
```

▶ **Scan Using Hostname**

```
nmap example.com
```

▶ **Aggressive Scan (OS + Version + Scripts)**

```
nmap -v -A scanme.nmap.org
```

- ✓ Finds open ports
- ✓ OS detection
- ✓ Service detection
- ✓ Script scanning (NSE)
- ✓ Fast & reliable

Nmap's official practice target:

scanme.nmap.org

| ⚠ Important: Scanning real networks without permission is illegal.

◆ Nikto (Web Server Vulnerability Scanner)

Nikto scans web servers for:

- Old software versions
- Dangerous files
- Misconfigurations
- Outdated Apache modules

► Basic Nikto Scan

```
nikto -host scanme.nmap.org
```

Results show weaknesses such as:

- ✓ Old server versions
- ✓ Missing security headers
- ✓ Directory indexing enabled
- ✓ Vulnerable modules

These insights help testers plan exploitation.

◆ Nessus (Professional Vulnerability Scanner)

Nessus is one of the **industry-leading vulnerability scanners**.

✓ Features:

- Deep vulnerability scans
- CVE detection
- Risk scoring
- Compliance checks
- Detailed reports

► Installation Steps (Simplified)

1. Download from:

<https://www.tenable.com/downloads/nessus>

2. Install using:

```
dpkg -i Nessus.deb
```

1. Start service:

```
/etc/init.d/nessusd start
```

1. Access UI:

```
https://kali:8834/
```

1. Enter activation code
2. Create a **Basic Network Scan**
3. Start the scan
4. View vulnerabilities under "Reports"

Nessus delivers:

- ✓ High-level summaries
- ✓ Critical/High/Medium/Low ratings
- ✓ Remediation recommendations
- ✓ CVE numbers

◆ Service Enumeration

After finding open ports, the next step is **enumeration**, such as:

SSH Enumeration:

```
nmap -sV --script=ssh* target
```

SMB Enumeration:

```
enum4linux -a target
```

DNS Zone Transfer Check:

```
dig @target.com axfr
```

◆ Banner Grabbing

Attackers often extract service banners to identify software versions.

Example:

```
nc -v target 80
```

◆ Firewall & IDS Evasion Techniques

Nmap supports techniques to bypass firewalls:

- Fragmentation (`f`)
- Decoys (`D`)
- Mac Spoofing (`-spoof-mac`)
- Timing options (`T1` , `T2`)

◆ Why Network Scanning Matters in Cybersecurity

Network scanning is one of the **core skills** for:

- Penetration Testers
- SOC Analysts
- Red Teamers
- Bug hunters

- System Admins
- Security Engineers

It helps answer:

- What services are running?
- Which ports are exposed?
- Which versions are outdated?
- What are my attack surfaces?
- What vulnerabilities exist?

The more information gathered, the easier and more accurate the attack simulation.

Important Ethical Warning

Unauthorized scanning of networks is:

- ✗ Illegal
- ✗ Traceable
- ✗ Punishable

Always use:

- ✓ Lab machines
 - ✓ Your own network
 - ✓ Approved targets
 - ✓ Educational platforms (TryHackMe, HTB)
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Top Scanning Mistakes Beginners Make

- Scanning without permission
- Using only basic scans
- Ignoring version detection

- Forgetting UDP scans
 - Not correlating results
 - Not analyzing returned banners
 - Blindly trusting vulnerability scanners
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Final Summary

Network scanning is **not just a step** — it is the **foundation** of hacking and defense.

- ✓ Reconnaissance = Knowing the target
- ✓ Scanning = Finding attack surfaces
- ✓ Nmap = Port & service scanning
- ✓ Nikto = Web server scanning
- ✓ Nessus = Deep vulnerability assessment

Combined, they give a complete picture of a target's weaknesses and help cybersecurity teams secure systems proactively.
